

F215 Digital Design — Tutorial 1

Working in Binary: representations, complements, Gray code, Boolean algebra

Question 1 — Representation conversions

A weather station's temperature sensor gives its reading directly as an 8-bit binary whole number. This value is fed to two places: a BCD display module which accepts 8421-BCD encoded digits for display, and a debug tool which displays the value in hexadecimal format. Later in the design, this same value also needs a 9's-complement operation performed on it as part of a subtraction elsewhere in the circuit.

- (a) How many BCD digits will the display module require?
- (b) The sensor's current reading is 10101111. What is this value in decimal?
- (c) What value will the debug tool display?
- (d) What binary input will you feed to the 8421-BCD display module?
- (e) Given the later 9's-complement requirement, which digit encoding would you choose instead of plain 8421-BCD?

Question 2 — Gray code: mirror method, and a second method

Rotary shaft encoders report angular position using Gray code instead of plain binary, because in binary, transitions like 011→100 flip three bits at once — if even one bit is read a fraction late, the sensor can briefly report a completely wrong angle.

- (a) Using the mirror method, hand-derive the 4-bit Gray code sequence (16 rows), with columns $b_3b_2b_1b_0$ (binary) and $g_3g_2g_1g_0$ (Gray).
- (b) Pick any one row of your table. Make three columns: col 1 = $b_3b_2b_1b_0$, col 2 = $0b_3b_2b_1$ (shift right, pad with a leading 0), col 3 = $g_3g_2g_1g_0$. What pattern do you observe between the three columns? Try with a few more rows of the table from part (a). Does it correspond to some gate?
- (c) Repeat (b) for the reverse conversion — building the binary column from the Gray column. What pattern do you observe this time?
- (d) An encoder manufacturer wants to build the binary→Gray conversion directly into the sensor's output wiring, converting each reading in real time. Which of the two methods — mirroring, or the rule from (b) — is more suitable for such a sensor? Give our reasoning.

Question 3 — Why 2's-complement makes arithmetic “just work”

- (a) Take $A = 13$ and $B = -9$. Write both in 6-bit signed-magnitude, and separately in 6-bit 2's-complement. In each system, try computing $A + B$ by directly adding the two bit patterns as if they were plain unsigned binary numbers. What happens in each case — does the raw addition give you the correct answer, correctly encoded?
- (b) Repeat for $A = -15$ and $B = -6$.
- (c) Based on (a) and (b), explain what “just works” with 2's-complement addition that doesn't work the same way with signed-magnitude, and why.
- (d) Suppose your calculator circuit only knows how to add — no separate subtraction circuit. Using what you found above, explain how you'd get it to compute $A - B$ for two 6-bit signed numbers anyway.

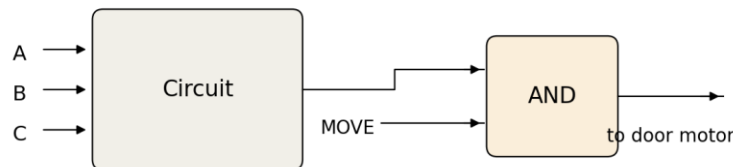
Question 4 — From requirements to a simplified expression

A door safety circuit is wired as shown below, with inputs A, B, C feeding a logic circuit whose output goes into an AND gate alongside a MOVE signal from the rest of the system:

$A = 1$ if the door is fully closed

$B = 1$ if the motion sensor path is clear

$C = 1$ if manual override is active



The overall system should let the door move in exactly these cases: whenever the door is closed and the path is clear; OR whenever the door is not closed, but either the path is blocked or override is off; OR whenever the door is closed but the path is blocked and override is active.

(a) What must the circuit box compute, given this description? Write it as a Boolean expression $F(A,B,C)$, then simplify it using Boolean algebra, showing each law you use.

(b) Should the circuit's output — the signal feeding the AND gate — be F , or F' ? Derive the correct expression two ways: (i) De Morgan's laws applied to your simplified F , (ii) the duality principle. Confirm both match.